A Contrario Paradigm for YOLO-Based Infrared Small Target Detection Analysis and Reproduction

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1 Introduction

Infrared small target detection is difficult because the targets often occupy only a few pixels, while the background contains clouds, sensor noise, and textured structures that can locally look similar to true targets. In that setting, the main challenge is often not to propose boxes, but to avoid false alarms. This is the point addressed by the paper *A Contrario Paradigm for YOLO-Based Infrared Small Target Detection*, which replaces the usual YOLO objectness estimation by an a contrario score measuring whether the content of a predicted box is too concentrated to be explained by a naive background model [1]. However, the paper alone does not provide sufficient implementation detail to fully reproduce the method. Additional information from the first author's thesis [2] was required to clarify several aspects of the model architecture and training procedure. In particular, the thesis reveals that the authors introduced ~ 400K additional parameters on top of the ~ 6M parameters of the baseline YOLOv7-tiny model.

In this report, we study this approach from the perspective of the course. We describe the detection problem, the naive model, the tested events, the Number of False Alarms (NFA), its validity, the search strategy, and the exclusion principle. We also detail the detector implemented in our repository. Beyond reproducing the object-level OL-NFA head described in the paper, our implementation includes practical additions such as a score-adapter convolution per detection scale and an ECA-style attention block that reweights the three significance maps before converting them into objectness. These additions were designed to be consistent with the reported ~ 400K parameter increase, since the thesis does not provide a detailed breakdown of the added components. Code is available at https://github.com/eliasalbouzidi/yolo-nfa-small-target-detection.

The method can be summarized as follows. Standard YOLO learns an objectness score directly from convolutional features. In contrast, OL-NFA (Object-Level Number of False Alarms) evaluates whether the feature responses inside a predicted box are unusually concentrated relative to the background. A box receives a high objectness score when this concentration is statistically significant, meaning it is highly unlikely to arise by chance under the naive model. This interpretation is closely related to Gestalt theory and the Helmholtz principle: a structure is meaningful when it is too improbable to be accidental.

2 Overview of the Method

2.1 Standard YOLO

The baseline detector is YOLOv7-tiny [3]. Starting from an input image of size 640×640, the backbone and the neck apply a sequence of convolutional layers that progressively reduce the spatial resolution while increasing the number of channels. The idea is to compress the image in the spatial dimensions while building richer feature representations. After this stage, the detector outputs three feature maps:

$$F_1 : \ 80 \times 80 \times 128, \qquad F_2 : \ 40 \times 40 \times 256, \qquad F_3 : \ 20 \times 20 \times 512.$$

These three maps correspond to three different strides with respect to the input image:

$$s_1 = 8, \qquad s_2 = 16, \qquad s_3 = 32.$$

This means that one cell of F_1 corresponds to an 8×8 region in the input image, one cell of F_2 corresponds to a 16×16 region, and one cell of F_3 corresponds to a 32×32 region. The finest map F_1 is therefore the most useful for very small targets, while F_2 and F_3 provide coarser but more contextual information.

On each of these feature maps, YOLO places a small set of anchors at every spatial location. An anchor is a predefined reference box shape attached to one grid cell. In YOLOv7-tiny, there are $A = 3$ anchors per cell at each detection scale. At a given position (u, v) on a feature map, the detector does not test only one possible box there, but rather 3 candidate box templates with different sizes and aspect ratios.

This is the purpose of anchors. A single grid cell may correspond to several possible object shapes, and the anchors provide these initial geometric hypotheses. The network then predicts corrections with respect to each anchor, rather than predicting all boxes completely from scratch. In practice, for each anchor at each cell, the detector predicts $(x, y, w, h, \text{obj}, \text{cls})$, that is four box parameters, one objectness logit, and one class logit.

The four box parameters should be understood as follows. The pair (x, y) tells the detector where the center of the predicted box is relative to the current grid cell, while (w, h) tells it how the anchor width and height should be adjusted. After decoding, each anchor therefore becomes one candidate bounding box in image coordinates.

This explains why the last layer of the YOLO head is a 1×1 projection convolution. It takes the local feature vector at one grid location and converts it into all anchor predictions attached to that location.

So the standard YOLO head works in two steps. First, the backbone and neck produce the feature maps F_1, F_2, and F_3. Second, one projection convolution per scale transforms each local feature vector into anchor-wise predictions. From these predictions, YOLO decodes a large set of candidate bounding boxes together with one objectness score per candidate.

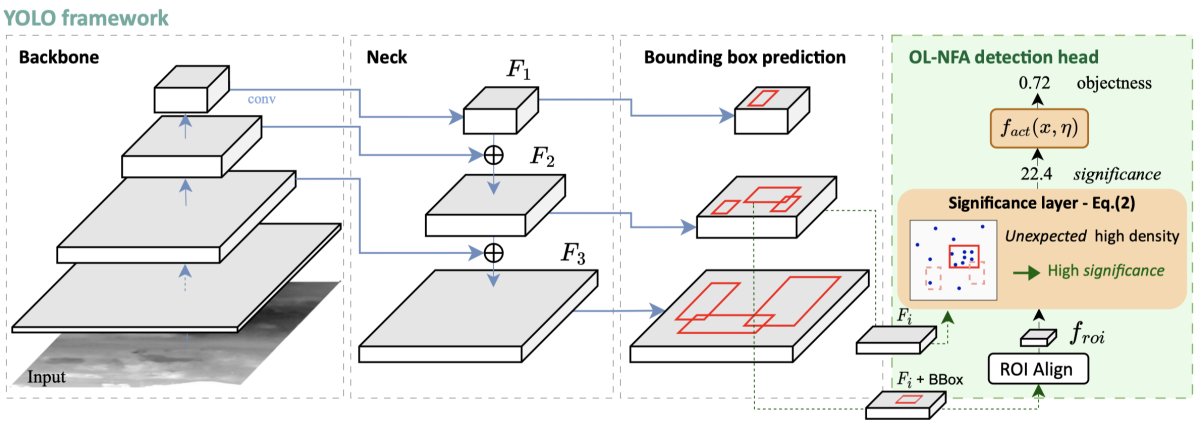


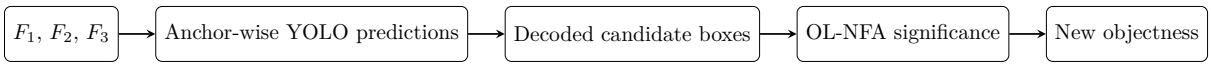
Figure 1: High-level overview of the YOLOv7-tiny + OL-NFA detector. The backbone and neck produce three detection feature maps. Bounding boxes are predicted as in standard YOLO, while the objectness score is re-estimated from an a contrario significance computed on ROI-aligned features.

2.2 YOLO OL-NFA

OL-NFA does not replace this whole mechanism. The anchors are still there, the projection convolutions are still there, and the bounding boxes are still decoded exactly as in YOLOv7-tiny.

What changes is only the objectness branch. Instead of using the original YOLO objectness logit, the detector recomputes objectness from an a contrario significance score.

The pipeline can therefore be summarized as follows.



More precisely, once YOLO has produced candidate boxes, each box is sent back to the corresponding feature map. ROI Align extracts the region covered by that box, OL-NFA evaluates whether the responses inside this region are more concentrated than expected under a background-only model, and this significance is converted into a new objectness score. In other words, YOLO still proposes where an object may be, and OL-NFA re-estimates how meaningful this proposal is.

3 A Contrario Formulation

3.1 Detection problem and naive model

Detection problem. Given an infrared image I, the goal is to output a non-redundant set of bounding boxes around true targets while rejecting accidental background structures. This can be interpreted in Gestalt terms, where a target should correspond to a meaningful structure, not to an accidental fluctuation of the background. In the present method, meaningful means statistically surprising under a naive model.

Naive model. The object-level formulation used here starts from a Bernoulli background model. On a single-channel map, each elementary position is assumed to be active independently with probability p. If a region has area ν, then the number of active points inside it is a binomial law under the null hypothesis H_0:

$$K \mid H_0 \sim \text{Binomial}(\nu, p).$$

The thesis clarifies how this formulation is extended to multi-channel feature maps, where the tested region becomes a 3D volume whose third axis is the channel dimension [2]. In practice, this means that the tested volume is

$$\nu = \text{ROI area on the feature map} \times \text{number of channels}.$$

3.2 Event family, precision parameters, and number of tests

The tested events are not arbitrary rectangles over the image. They are the boxes predicted by the YOLO head on the three detection scales:

$$P_3 : 80 \times 80, \qquad P_4 : 40 \times 40, \qquad P_5 : 20 \times 20.$$

With $A = 3$ anchors per location, the total number of candidate events per image is

$$\eta_{\text{test}} = 3(80 \cdot 80 + 40 \cdot 40 + 20 \cdot 20) = 25200.$$

One event is defined by the following precision parameters (i, a, u, v, B, C_i), where i is the detection scale, a is the anchor index, (u, v) is the grid location, $B = (x_1, y_1, x_2, y_2)$ is the decoded predicted box, and C_i is the number of channels of the corresponding feature map. The family of tests is therefore finite and easy to count, which is one of the requirements for defining an NFA.

3.3 NFA formula

Let κ be the observed number of active points inside a tested ROI of volume ν. Under the naive model, the meaningful event is $\{K \geq \kappa\}$, which corresponds to observing at least κ active points in a region where the background contains active points only with probability p.

Its tail probability under H_0 is

$$\mathbb{P}_{H_0}(K \geq \kappa) = \sum_{t=\kappa}^{\nu} \binom{\nu}{t} p^t (1-p)^{\nu-t}. \tag{1}$$

The corresponding Number of False Alarms is

$$\mathrm{NFA}(\kappa, \nu, p) = \eta_{\text{test}} \sum_{t=\kappa}^{\nu} \binom{\nu}{t} p^t (1-p)^{\nu-t}. \tag{2}$$

This follows the standard a contrario form [4]:

$$\mathrm{NFA} = \text{number of tests} \times \text{tail probability under } H_0.$$

Validity of the NFA. If one freezes the family of tested ROIs (regions of interest) and uses the discrete count κ, then Eq. (2) is a valid NFA in the classical sense, meaning that for every $\varepsilon > 0$, the expected number of false detections such that $\mathrm{NFA} \leq \varepsilon$ is at most ε. This is the usual guarantee recalled in the course. In the implemented detector, however, the tested boxes are predicted by the network and the discrete quantities are replaced by smooth approximations. The exact theorem therefore applies to the fixed-family formulation, while the detector used in practice should be understood as a differentiable surrogate built from the same a contrario principle.

3.4 Differentiable approximation used in practice

The exact formula in Eq. (2) cannot be inserted directly into the training loop for two reasons. First, the feature maps are real-valued rather than binary, so κ is not directly available. Second, the exact binomial tail is not convenient for stable backpropagation.

At detection scale i, let F_i denote the original feature map produced by the YOLO neck:

$$F_1 \in \mathbb{R}^{128\times80\times80}, \qquad F_2 \in \mathbb{R}^{256\times40\times40}, \qquad F_3 \in \mathbb{R}^{512\times20\times20}.$$

In the implemented detector, OL-NFA is not applied directly to these raw YOLO feature maps. We add a score-adapter block at each scale:

$$A_1 : \mathbb{R}^{128\times80\times80} \to \mathbb{R}^{64\times80\times80},$$

$$A_2 : \mathbb{R}^{256\times40\times40} \to \mathbb{R}^{64\times40\times40},$$

$$A_3 : \mathbb{R}^{512\times20\times20} \to \mathbb{R}^{64\times20\times20}.$$

Each block keeps the spatial resolution unchanged and maps the channel dimension to a common value 64. Each A_i is implemented as

$$\mathrm{Conv}_{1\times1}(C_i \to 64) \;\to\; \mathrm{Conv}_{3\times3}(64 \to 64) \;\to\; \mathrm{Conv}_{3\times3}(64 \to 64) \;\to\; \mathrm{Conv}_{1\times1}(64 \to 64).$$

The adapted feature map used by OL-NFA is therefore

$$\widetilde{F}_i = A_i(F_i) \in \mathbb{R}^{64\times H_i\times W_i}.$$

All OL-NFA statistics are computed on $\widetilde{F}_i$, not on the original F_i.

For each predicted box $B = (x_1, y_1, x_2, y_2)$, ROI Align extracts a tensor

$$R(B) = \mathrm{RoIAlign}(\widetilde{F}_i, B) \in \mathbb{R}^{64\times 5\times 5}.$$

The hard count is replaced by a soft count:

$$\kappa(B) = \sum_{c,u,v} \sigma(R(B)_{cuv}), \tag{3}$$

where σ is the sigmoid function. Each sample therefore contributes continuously between 0 and 1 instead of being thresholded.

The global background density at the same scale is estimated by

$$p_i = \frac{1}{64\, H_i W_i} \sum_{c,x,y} \sigma(\widetilde{F}_i(c, x, y)). \tag{4}$$

The local density inside the ROI is then

$$q(B) = \frac{\kappa(B)}{25 \cdot 64}. \tag{5}$$

The tested volume keeps the same interpretation as in the theoretical formulation:

$$\nu(B) = 64 \cdot \frac{(x_2 - x_1)(y_2 - y_1)}{s_i^2}, \tag{6}$$

where $s_i \in \{8, 16, 32\}$ is the stride of the corresponding scale.

Table 1: Correspondence between the theoretical NFA quantities and the implemented differentiable approximation.

Theoretical quantity	Implemented quantity
κ	soft count $\sum \sigma(R(B)_{cuv})$
p	global density p_i on the adapted feature map
ν	box area on the adapted feature map $\times 64$

Instead of working directly with the NFA, the authors introduce the significance

$$S(B) = -\ln(\mathrm{NFA}(B)).$$

The exact binomial tail is approximated by a Hoeffding–Chernoff large-deviation formula [4] when the local density exceeds the background density:

$$\mathbb{P}_{H_0}(K \geq \kappa) \approx \exp(-\nu(B)\, D_{\mathrm{KL}}(q(B) \| p_i)) \qquad \text{if } q(B) > p_i,$$

where

$$D_{\mathrm{KL}}(q \| p) = q \ln \frac{q}{p} + (1 - q) \ln \frac{1 - q}{1 - p}.$$

This leads to the practical approximation

$$S(B) = \begin{cases} \nu(B)\, D_{\mathrm{KL}}(q(B) \| p_i) - \ln \eta_{\mathrm{test}}, & q(B) > p_i, \\ -\ln \eta_{\mathrm{test}}, & q(B) \leq p_i. \end{cases} \tag{7}$$

The objectness probability finally used by the detector is obtained from the significance by

$$o(B) = \mathrm{clip}_{[0,1]}\left(2\,\sigma(\alpha_{\mathrm{sigm}}(S(B) + \ln \eta_{\mathrm{test}})) - 1\right), \qquad \alpha_{\mathrm{sigm}} = 0.5. \tag{8}$$

In our implementation, there is one additional step before this conversion. The three scale-wise significance maps are reweighted by an ECA-style attention block, so the practical sequence is

$$S_i(B) \;\longrightarrow\; w_i\, S_i(B) \;\longrightarrow\; o_i(B),$$

where w_i is a learned scale weight. The score-adapter and attention block are explained in Section 7.1.

3.5 Fast search strategy and exclusion principle

Fast search strategy. The detector does not scan all possible rectangles in the image. Instead, it uses YOLO itself as a proposal generator. Candidate boxes are obtained in one forward pass, and OL-NFA is computed only on these proposals. The search is therefore proposal-driven rather than exhaustive.

For an image of size 640×640, only 25200 candidate boxes are evaluated. This is much cheaper than a classical sliding-window a contrario search over all rectangles. The main additional cost comes from ROI Align and the significance computation. In practice, the code also processes ROIs by chunks in order to limit memory usage.

Possible improvements. From the perspective of the a contrario framework, the current implementation does not fully exploit classical fast evaluation strategies. In a purely classical setting, one would typically rely on a binarized score map together with integral images or integral histograms, allowing the efficient evaluation of a very large number of rectangular regions. However, this approach is not directly compatible with the present detector, which operates on multi-channel, real-valued feature maps and uses ROI Align.

More practically, the computational cost of the current method could be reduced by limiting the number of evaluated proposals. Instead of computing the OL-NFA score on all predicted boxes, one could first select a subset of the most promising candidates (for instance the top-k boxes according to the YOLO objectness score) and apply the NFA only to this reduced set. This would preserve the proposal-driven nature of the method while significantly lowering the cost of ROI extraction and significance computation.

Exclusion principle. The final non-redundant detections are produced with the standard YOLO post-processing, where low-score boxes are filtered, then Non-Maximum Suppression removes overlapping detections. From the point of view of the course, NMS plays the role of the exclusion principle, since it keeps the strongest detection in a local cluster and suppresses redundant hypotheses.

4 Implemented Detector and Training Protocol

4.1 Exact computation performed by the head

For each detection scale, the implemented head performs the following operations. First, the standard YOLO prediction convolution produces raw anchor predictions. Second, the bounding boxes are decoded in image coordinates with the usual YOLOv7 formulas. Third, the corresponding feature map is optionally transformed by the score adapter. Fourth, ROI Align

extracts a tensor of shape $C_i \times 5 \times 5$ for every decoded box. Fifth, the significance is computed with Eq. (7). Sixth, the three scale-wise significance maps are reweighted by the ECA-style block. Seventh, the reweighted significance is converted into an objectness probability through Eq. (8). Finally, that probability is turned back into a logit and replaces the original YOLO objectness logit.

The box regression branch is unchanged. OL-NFA only replaces the objectness branch.

4.2 ROI Align

We use the `torchvision` ROI Align operator with

$$\mathrm{RoIAlign}(\widetilde{F}_i, B,\ \mathrm{output_size} = 5,\ \mathrm{sampling_ratio} = 2,\ \mathrm{spatial_scale} = 1/s_i),$$

and `aligned=False` [5]. Concretely, the predicted box is expressed in image coordinates, then mapped to feature-map coordinates through the factor $1/s_i$. The corresponding box is divided into a 5×5 grid, each bin is sampled at 2×2 locations, the values at those locations are obtained by bilinear interpolation, and the four sampled values are averaged.

This operator is useful here because it preserves continuous box coordinates, which is important for stable backpropagation through the predicted boxes.

4.3 Loss and target assignment

The object-level OL-NFA head does not introduce a completely new detection loss. Once the significance has been converted into an objectness probability in $[0, 1]$, the detector is trained with the standard YOLO objectness loss, as explained in the thesis [2]. In our repository, the training loop uses the YOLOv7 loss.

The total loss is

$$\mathcal{L} = \lambda_{\mathrm{box}}\mathcal{L}_{\mathrm{box}} + \lambda_{\mathrm{obj}}\mathcal{L}_{\mathrm{obj}} + \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{cls}}, \tag{9}$$

with

$$\lambda_{\mathrm{box}} = 0.05, \qquad \lambda_{\mathrm{obj}} = 1.0, \qquad \lambda_{\mathrm{cls}} = 0.5.$$

The important point is the objectness term. In plain YOLO, the network predicts an objectness logit directly. In OL-NFA, the network first predicts a box, then OL-NFA computes a significance, converts it into an objectness probability through Eq. (8), and finally converts it back into a logit. This new logit replaces the original YOLO objectness logit. The objectness loss is then the usual BCE-with-logits loss:

$$\mathcal{L}_{\mathrm{obj}} = \sum_{i=1}^{3} \beta_i \,\mathrm{BCEWithLogits}(z_{\mathrm{obj}}^{(i)}, t_{\mathrm{obj}}^{(i)}), \tag{10}$$

with scale weights

$$\beta = (4.0, 1.0, 0.4)$$

.

For matched predictions, the target is the IoU itself:

$$t_{\mathrm{obj}} = \mathrm{IoU}(g_j, p_k),$$

and for unmatched predictions the target is 0. On the NUAA-SIRST dataset used here, there is only one class, so the classification term is inactive in practice:

$$\mathcal{L}_{\mathrm{cls}} = 0.$$

5 Dataset, Examples, and Experimental Protocol

5.1 Dataset

Following the paper, the main experiments are carried out on the NUAA-SIRST dataset [6]. This dataset contains 427 infrared images acquired in the wavelength range from 950 to 1200 nm. The targets are very small, often only a few pixels wide, and appear in challenging backgrounds such as clouds, clutter, and sensor noise. We use a 60:20:20 split for train, validation, and test, and also evaluates few-shot settings with only 25 training images.

This dataset is well suited to OL-NFA because it is a difficult false-alarm setting, since the targets are tiny and the background can easily generate compact bright structures.

Figure 2 shows 16 sample images from the NUAA-SIRST test set with ground-truth annotations. Several backgrounds such as uniform sky and smooth clouds are close to the Bernoulli naive model: the pixel activations appear roughly independent and uniformly distributed. In these scenes, a small target stands out as a compact bright cluster that is statistically improbable under H_0, which is precisely the setting where the a contrario test is most effective. Other images have structured backgrounds that locally violate the independence assumption and are therefore more challenging for the NFA formulation.

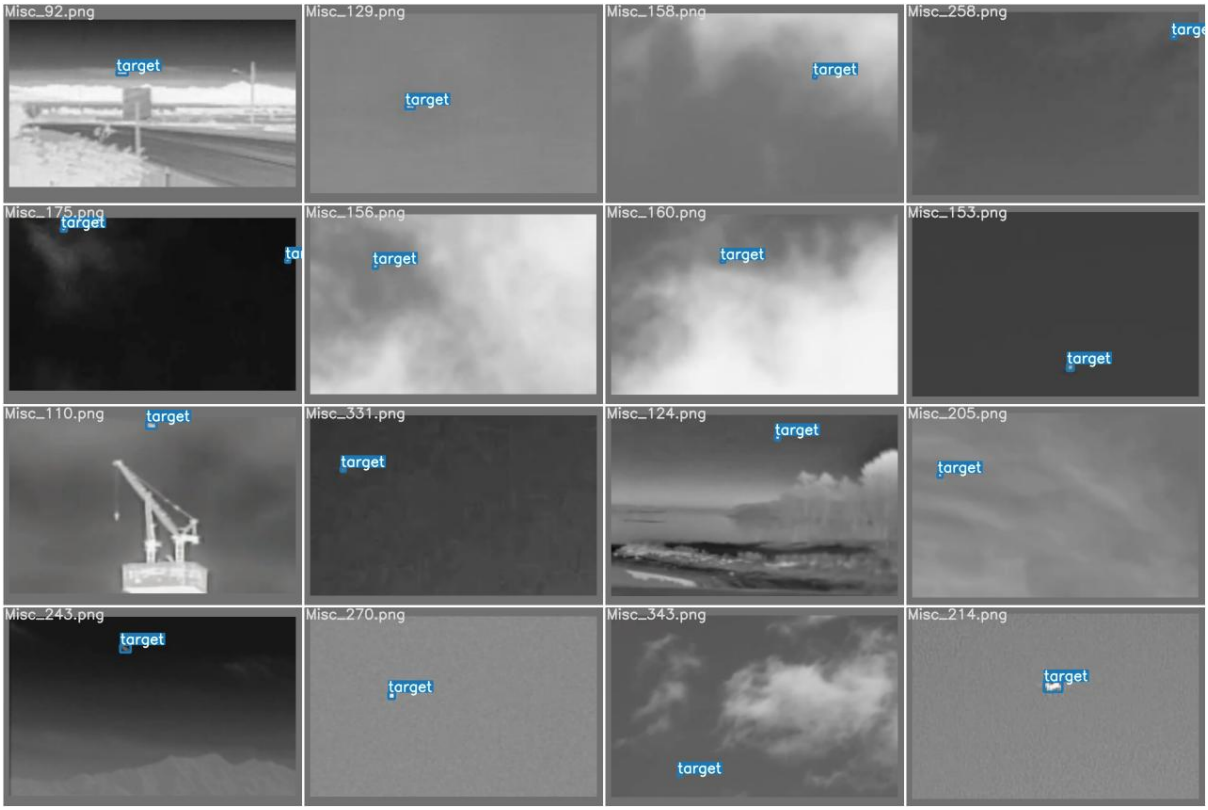


Figure 2: Sample images from the NUAA-SIRST test set with ground-truth annotations.

Figure 3 shows the predictions of the OL-NFA detector on the same 16 test images. Almost all targets are correctly detected with confidence scores ranging from 0.3 to 0.8, but some false alarms are produced. The confidence scores remain moderate rather than extreme, reflecting the difficulty of the task and the small size of the targets. Overall, the detector captures most meaningful structures while still producing a limited number of spurious detections, illustrating the expected trade-off between sensitivity and false alarm control.

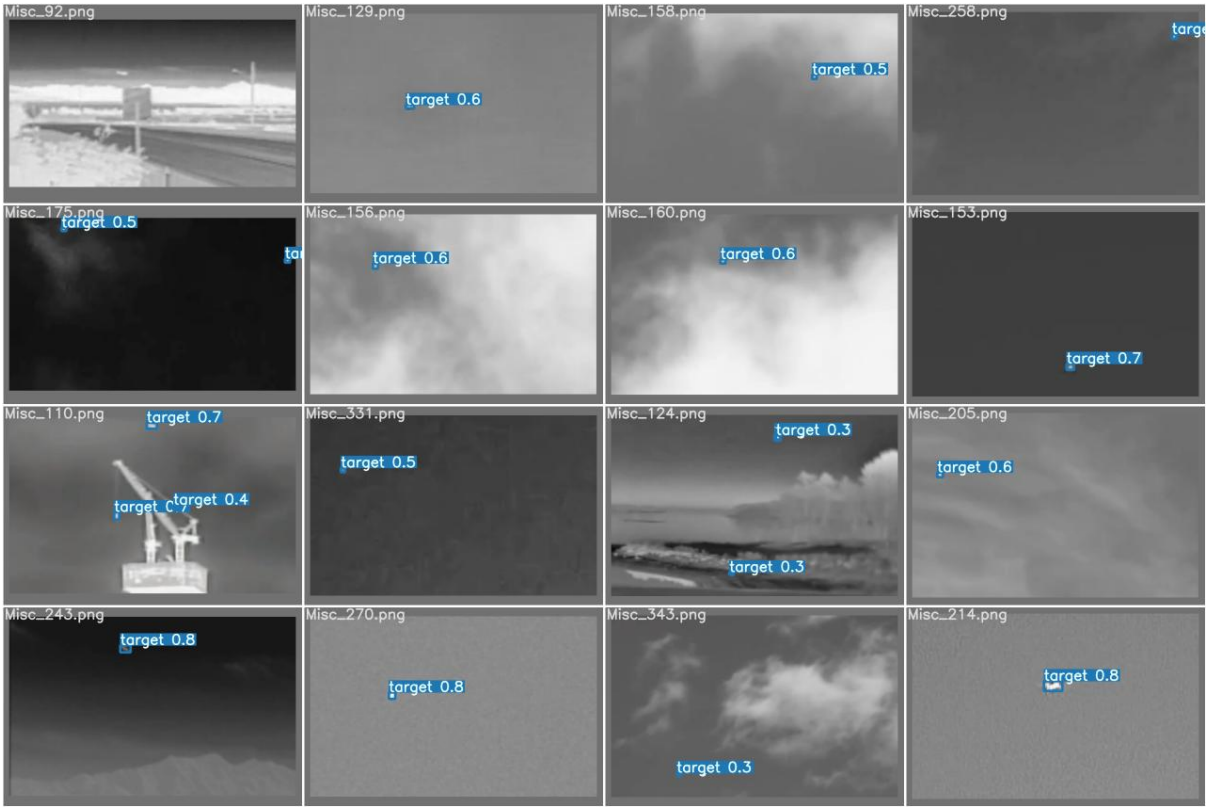


Figure 3: Predictions of the OL-NFA detector on the same test images.

5.2 Experimental protocol

To stay aligned with the paper, we compare the two detectors below under the same overall protocol.

Table 2: Training protocol used for the reproduction.

Setting	Value
Input size	640×640
Optimizer	Adam
Learning rate	10^{-3}
Batch size	16
Epochs	600
Training mode	from scratch
Compared models	YOLOv7-tiny / YOLOv7-tiny + OL-NFA

For the OL-NFA model, the implementation additionally uses the parameters listed in Table 3.

Table 3: OL-NFA hyperparameters.

Parameter	Value
`roi_size`	5
`sampling_ratio`	2
`sigm_alpha`	0.5
`score_adapter`	True
`scale_attention`	True

6 Results

Evaluation metrics. We report standard object detection metrics including precision, recall, F1-score, and average precision (AP). Due to the extremely small size of the targets (often only a few pixels), we evaluate detections using a relaxed Intersection over Union (IoU) threshold of 5%. In this regime, even a slight spatial misalignment between the predicted box and the ground truth can drastically reduce the IoU, potentially collapsing it to near zero. Therefore, a low IoU threshold is necessary to obtain a meaningful evaluation that reflects detection capability rather than sensitivity to minor localization errors.

Effect of OL-NFA. The results confirm the expected trade-off of an a contrario objectness criterion. The NFA formulation is designed to control the expected number of false detections under the background model, which naturally translates into higher precision and fewer false alarms. However, it also raises the detection bar, since targets whose feature activation is only slightly above the background density p_i will receive low significance and may be missed. This precision–recall trade-off is especially visible in the few-shot regime where the network has fewer examples to learn discriminative features.

6.1 Full dataset

Table 4 compares the paper's reported results with our reproduction. On the full training set, the baseline YOLOv7-tiny already performs well. In our experiments, OL-NFA does not improve the overall F1 or AP, since recall decreases from 92.7% to 89.7%, while precision remains unchanged. The gap can be attributed to implementation differences, since our version uses a score-adapter convolution and ECA-style scale attention that are not detailed in the original paper, which may use a slightly different hyperparameter configuration.

Table 4: Main NUAA-SIRST object-level results (full training set). Paper values from [1]; reproduction values from the present implementation.

	Paper			Reproduction		
Method	F1	AP	Prec./Rec.	F1	AP	Prec./Rec.
YOLOv7-tiny	96.5	97.8	96.9 / 96.2	**94.4**	**96.2**	96.2 / **92.7**
YOLOv7-tiny + OL-NFA	97.2	98.2	98.6 / 95.9	92.8	91.8	**96.2** / 89.7

6.2 Few-shot regime

Table 5 shows the results with only 25 training images. This is the setting where the a contrario principle provides the most interpretable benefit. Although OL-NFA does not improve F1 or AP overall, the precision increases significantly (79.5 → 85.5, i.e. +6 percentage points) and the number of false alarms per image is nearly halved (0.209 → 0.116). The cost is a recall drop (64.2 → 54.1), as the detector becomes more conservative.

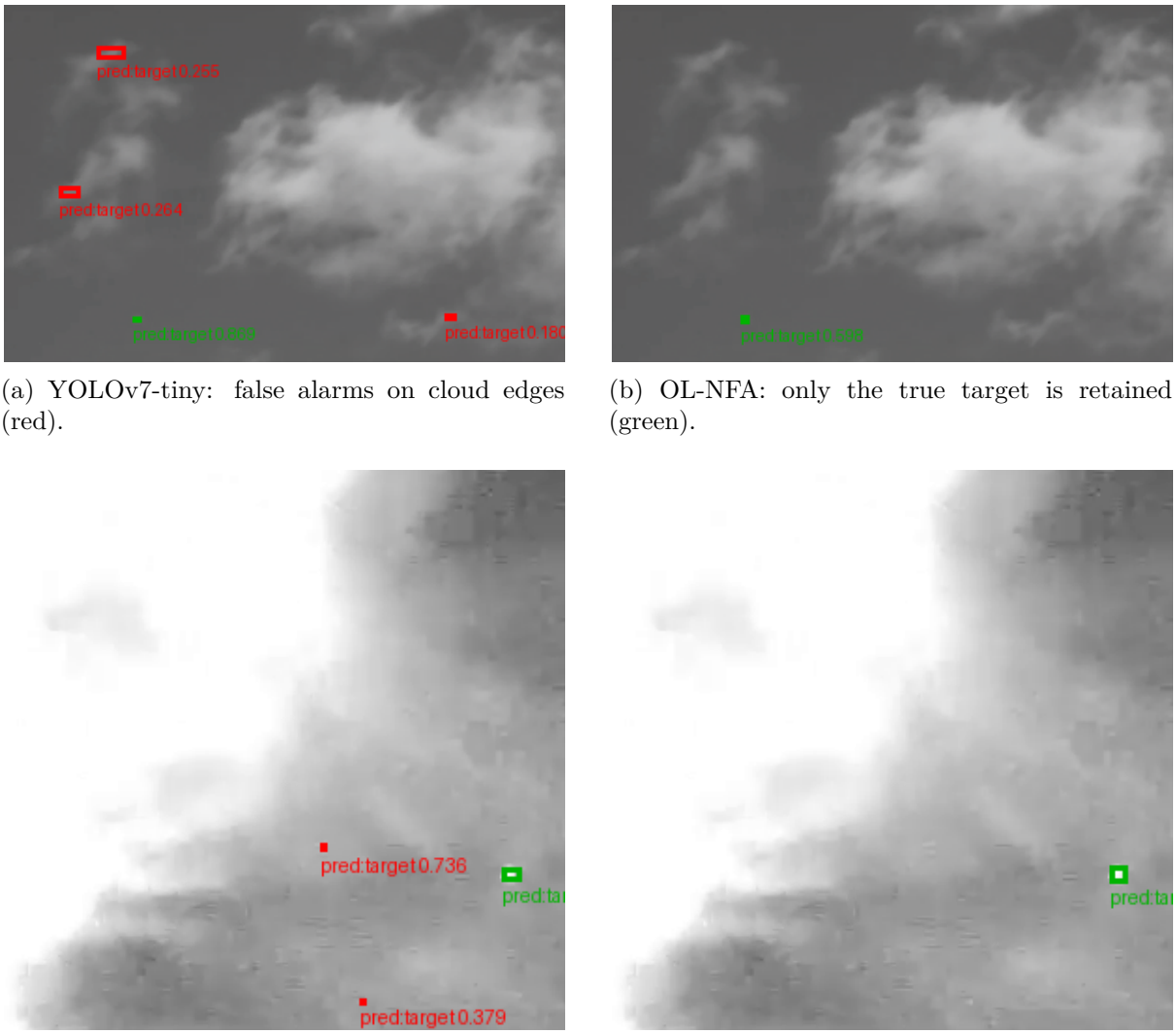
Table 5: Few-shot NUAA-SIRST results (25 training images). Bold indicates the best value per metric.

Method	F1	AP	Precision	Recall	FA/image
YOLOv7-tiny	**71.1**	**65.3**	79.5	**64.2**	0.209
YOLOv7-tiny + OL-NFA	66.3	59.5	**85.5**	54.1	**0.116**

OL-NFA systematically improves precision and reduces false alarms at the expense of recall.

Figure 4 shows two representative examples trained with only 25 images. In both cases, the YOLOv7-tiny baseline produces false positives on cloud edges or textured regions (red boxes), while the OL-NFA detector removes them and retains only the true target (green boxes). Background structures that are locally bright but statistically consistent with H_0 receive low significance and are correctly rejected.

The main benefit of OL-NFA appears in the few-shot setting, where limited training data makes the baseline detector more prone to false alarms.



(a) YOLOv7-tiny: false alarms on cloud edges (red).

(b) OL-NFA: only the true target is retained (green).

(c) YOLOv7-tiny: false alarms in cluttered background.

(d) OL-NFA: false positives removed.

Figure 4: Complex cases from the few-shot regime (25 training images). Left column: YOLOv7-tiny baseline with false positives in red. Right column: YOLOv7-tiny + OL-NFA removes the false alarms while keeping the true detection.

7 Discussion and Conclusion

7.1 Difference with the original paper

The original paper describes the OL-NFA formulation in a 2D setting, where the tested regions are defined only over the spatial support of the feature map. In that formulation, the quantity

ν corresponds to the area of the region, and the naive model is implicitly applied on a single-channel map [1]. However, in practice, the feature maps F_i produced by the YOLO backbone are multi-channel tensors of size $H_i \times W_i \times C_i$. This discrepancy is not explicitly discussed in the paper.

The thesis clarifies this point by introducing a multi-channel interpretation of the object-level formulation, where the tested region becomes a 3D volume rather than a 2D area. The third dimension corresponds to the channel axis, leading to:

$$\nu(B) = \text{ROI area on the feature map} \times C_i.$$

Our implementation follows this interpretation and therefore relies on a 3D NFA formulation, where the statistical test is performed over a volume of activations instead of a planar region.

A direct consequence of this multi-channel setting is that the feature maps at different scales have different channel dimensions (128, 256, and 512 channels). In our implementation, OL-NFA is therefore not applied directly to the raw YOLO maps. A score-adapter block is inserted before the NFA computation at each scale. Concretely, the implemented architecture is

$$\begin{aligned} 80 \times 80 \times 128 &\longrightarrow 80 \times 80 \times 64, \\ 40 \times 40 \times 256 &\longrightarrow 40 \times 40 \times 64, \\ 20 \times 20 \times 512 &\longrightarrow 20 \times 20 \times 64. \end{aligned}$$

Each adapter is a small convolutional block:

$$\text{Conv}_{1\times1}(C_i \to 64) \;\to\; \text{Conv}_{3\times3}(64 \to 64) \;\to\; \text{Conv}_{3\times3}(64 \to 64) \;\to\; \text{Conv}_{1\times1}(64 \to 64).$$

As a result, the ROI features, the soft count $\kappa(B)$, the background density p_i, and the tested volume $\nu(B)$ are all computed on homogeneous 64-channel adapted maps. This score adapter is therefore a core part of the implemented OL-NFA head.

In addition, our implementation includes an engineering component that is not described in the paper: a scale attention mechanism inspired by Efficient Channel Attention (ECA). Concretely, this module assigns a learned weight w_i to each detection scale, allowing the network to modulate the relative importance of the different feature levels:

$$S_i(B) \;\longrightarrow\; w_i\, S_i(B).$$

This mechanism gives different importance to each scale and can be interpreted as a form of cross-scale calibration.

It is also worth noting that the thesis reports an increase of approximately 400K parameters on top of the $\sim$ 6M parameters of the baseline YOLOv7-tiny model [2], without providing a detailed breakdown of these additional components. The architectural choices introduced in our implementation, namely the score-adapter convolutions and the scale attention mechanism, are consistent with this order of magnitude ($\sim 400k$ parameters).

Overall, due to these modifications, namely the explicit 3D formulation, the introduction of score-adapter convolutions, and the scale attention mechanism, our implementation should be understood as a paper inspired extension of the object-level OL-NFA idea, rather than a strict reproduction of the original method.

7.2 Experimental findings

The experiments highlight distinct behaviors depending on the training regime.

On the full NUAA-SIRST dataset with sufficient training data, the OL-NFA module in our implementation does not improve upon the baseline in terms of F1 or AP. The precision is

preserved (96.2% for both), but recall drops from 92.7% to 89.7%. This indicates that, in this regime, the standard YOLO objectness is already well calibrated, and the additional a contrario criterion tends to be overly conservative, rejecting some true positives.

In contrast, in the few-shot regime (25 training images), OL-NFA provides a clear benefit in false alarm control. The false alarm rate is nearly halved ($0.209 \rightarrow 0.116$ per image), and precision improves by 6 percentage points ($79.5 \rightarrow 85.5$). This behavior is consistent with the theoretical motivation: when training data is scarce, the learned objectness becomes noisy, and the background-based statistical model acts as a regularizer that filters out detections that are not sufficiently meaningful.

Overall, these results show that the OL-NFA module mainly improves robustness in low-data regimes by enforcing a stricter statistical criterion. At the same time, it provides an interpretable objectness score grounded in the a contrario theory, where each detection is evaluated through its associated Number of False Alarms (NFA), making the decision process explicitly linked to statistical significance.

7.3 Limitations and possible improvements

The main theoretical limitation is that the exact NFA guarantee is strongest for a fixed finite family of tests with discrete counts. The implemented detector instead uses predicted boxes, soft counts, ROI sampling, and a KL-based approximation. It therefore keeps the spirit and the structure of the a contrario formulation, but it is more accurate to describe it as a differentiable significance module than as a fully classical NFA detector.

A practical limitation already discussed in the thesis is that the first object-level formulation tends to favor compact dense boxes, which can bias the detector toward smaller boxes and reduce localization quality on larger objects [2]. This is one reason why the thesis later introduced a second object-level formulation. A limitation specific to our reproduction is the sensitivity to the background density estimate p_i. Since p_i is computed as a global mean over the entire feature map (Eq. 4), it does not account for spatial variations in the background. In images where part of the background is bright (e.g. cloud edges) and part is dark, a global p_i may overestimate the background density in dark regions and underestimate it in bright regions, leading to both missed detections and false alarms.

Several directions could be explored to address these limitations. A natural extension would be to make the background density p_i spatially adaptive rather than global, in order to better capture local variations of the background. It would also be relevant to design an NFA formulation that directly exploits the multi-channel structure of the feature maps, instead of reducing them to scalar activations through the sigmoid function. Finally, reducing the computational cost could be achieved by introducing a more aggressive pruning of candidate boxes before applying ROI Align, so that the NFA is evaluated only on the most promising proposals.

7.4 Conclusion

This work illustrates how an a contrario principle can be integrated into a modern object detection framework. The underlying NFA formulation is simple and interpretable, as it combines a finite number of tests with a binomial tail probability under a background model. In practice, this exact formulation is replaced by a differentiable approximation, allowing the criterion to be embedded into the YOLO training pipeline.

The implemented detector preserves the YOLOv7-tiny architecture for bounding box regression and modifies only the objectness branch. It follows the object-level OL-NFA formulation clarified in the thesis, while incorporating additional architectural components such as score-adapter convolutions and scale-wise attention. These elements ensure consistency across multi-scale,

multi-channel feature maps and enable stable optimization.

The experimental results show that the OL-NFA module primarily improves false alarm control, especially in low-data regimes. In the few-shot setting, it significantly increases precision and reduces the false alarm rate, while in the full-data regime it leads to a decrease in recall. This behavior reflects the intrinsic nature of the a contrario framework, which favors statistically meaningful detections and penalizes ambiguous ones.

Overall, this approach provides an interpretable objectness score grounded in NFA theory, where each detection is explicitly associated with a measure of statistical significance relative to the background model.

Use of AI Tools

AI tools were used only as auxiliary support for optimization of the training pipeline and for language revision to improve the clarity and readability of the report.

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